

NCERT Exemplar Solutions for Class 12

Chemistry

Chapter 6- General Principles and Processes of Isolation of Elements

Multiple Choice Questions (Type-I)

1. In the extraction of chlorine by electrolysis of brine

- (i) oxidation of Cl^- ion to chlorine gas occurs.**
- (ii) reduction of Cl^- ion to chlorine gas occurs.**
- (iii) For overall reaction ΔG° has negative value.**
- (iv) a displacement reaction takes place.**

Ans: Option (i)

Explanation: $2\text{H}_2\text{O} + \text{Cl}^- \rightarrow 2\text{OH}^- + \text{H}_2 + \text{Cl}_2$

The ΔG° is +422 kJ for this reaction. We get $\Delta E^\circ = -2.2 \text{ V}$, when we convert it to ΔE° (using $G^\circ = E^\circ F$). It will, of course, require an external e.m.f. greater than 2.2 V. However, electrolysis necessitates an excess potential in order to overcome some other impeding reactions. Thus, Cl_2 is obtained through electrolysis, which produces H_2 and aqueous NaOH as byproducts. In addition, molten NaCl is electrolyzed. However, in that case, Na metal is formed rather than NaOH .

2. When copper ore is mixed with silica, in a reverberatory furnace copper matte is produced. The copper matte contains

- (i) sulphides of copper (II) and iron (II)**

(ii) sulphides of copper (II) and iron (III)

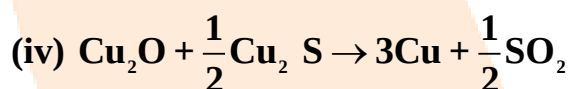
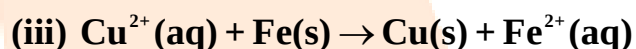
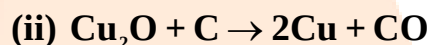
(iii) sulphides of copper (I) and iron (II)

(iv) sulphides of copper (I) and iron (III)

Ans: Option (iii)

Copper ore is mixed with silica before heating in a reverberatory furnace. When heated in a reverberatory furnace, the iron oxide slags of iron silicate and copper are produced. This product is produced in the form of copper matte in the furnace. Presence of Cu_2S and FeS is observed in copper matte. In these compounds, we observe the sulphides of Copper (I) Cu^{2+} and Iron (II) Fe^{2+} .

3. Which of the following reactions is an example of autoreduction?



Ans: Option (iv)

Copper (I) oxide is reduced by copper (I) sulphide in this reaction. Because copper is reduced by itself in this process, it is referred to as auto reduction.

4. A number of elements are available in earth's crust but most abundant elements are

(i) Al and Fe

(ii) Al and Cu

(iii) Fe and Cu

(iv) Cu and Ag

Ans: Option (i)

Explanation: Aluminium is the third most abundant metal in the earth's crust (8.3 percent approx. by weight). It is found in a variety of igneous minerals, including mica and clays. The second most abundant metal in the earth's crust is iron.

5. Zone refining is based on the principle that

(i) impurities of low boiling metals can be separated by distillation.

(ii) impurities are more soluble in molten metal than in solid metal.

(iii) different components of a mixture are differently adsorbed on an adsorbent.

(iv) Vapours of volatile compounds can be decomposed in pure metal.

Ans: Option (ii)

Purification of metal crystals by making a thin region of crystal undergo melting is known as Zone refining. The molten crystal is then moved up along the crystal to get pure form of it. This process is used to get pure form of Silicon and Germanium. Basic principle of the zone refining process is that the impurities are more soluble in molten metal than solid metal to get pure form of metal.

6. In the extraction of copper from its sulphide ore, the metal is formed by the reduction of Cu_2O with

(i) FeS

(ii) CO

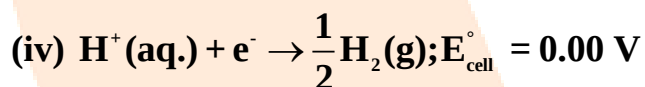
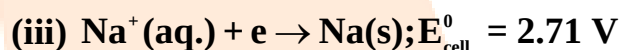
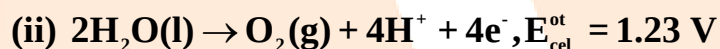
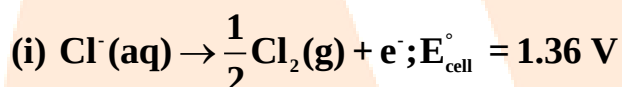
(iii) Cu_2S

(iv) SO_2

Ans: Option (iii)

Reaction takes place as $\text{Cu}_2\text{O} + \frac{1}{2}\text{Cu}_2\text{S} \rightarrow 3\text{Cu} + \frac{1}{2}\text{SO}_2$ with a product as bristle copper. This type of reaction is called an auto-reduction reaction as copper is reducing itself with help of other copper compounds.

7. Brine is electrolysed by using inert electrodes. The reaction at anode is



Ans: Correct option (i)

The reaction at the anode with a lower E° value is preferred, but oxygen cannot be obtained in this process due to overvoltage.

8. In the metallurgy of aluminium

(i) Al^{3+} is oxidised to $\text{Al}(\text{s})$.

(ii) graphite anode is oxidised to carbon monoxide and carbon dioxide.

(iii) oxidation state of oxygen changes in the reaction at anode.

(iv) oxidation state of oxygen changes in the overall reaction involved in the process.

Ans: Option (ii)

Reaction involved in the metallurgy of aluminium is $2\text{Al}_2\text{O}_3 + 3\text{C} \rightarrow 4\text{Al} + 3\text{CO}_2$.

The reaction at cathode is $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$

The reaction at anode is



Hence, from the reaction graphite anode is oxidised to carbon monoxide and carbon dioxide.

9. Electrolytic refining is used to purify which of the following metals?}

(i) Cu and Zn

(ii) Ge and Si

(iii) Zr and Ti

(iv) Zn and Hg

Ans: Option (i)

The electrolytic method can be used to purify zinc and copper. The impure metal is used as an anode in this method. As the cathode, a pure strip of the same metal is used. They are immersed in an appropriate electrolytic bath containing a soluble salt of the same metal.

The more basic metals remain in the solution, while the less basic metals are transferred to the anode mud.

10. Extraction of gold and silver involves leaching the metal with CN^- ion. The metal is recovered by

(i) displacement of metal by some other metal from the complex ion.

(ii) Roasting of metal complexes.

(iii) calcination followed by roasting.

(iv) thermal decomposition of metal complexes.

Ans: Option (i)

The cyanide process involves 3 steps:

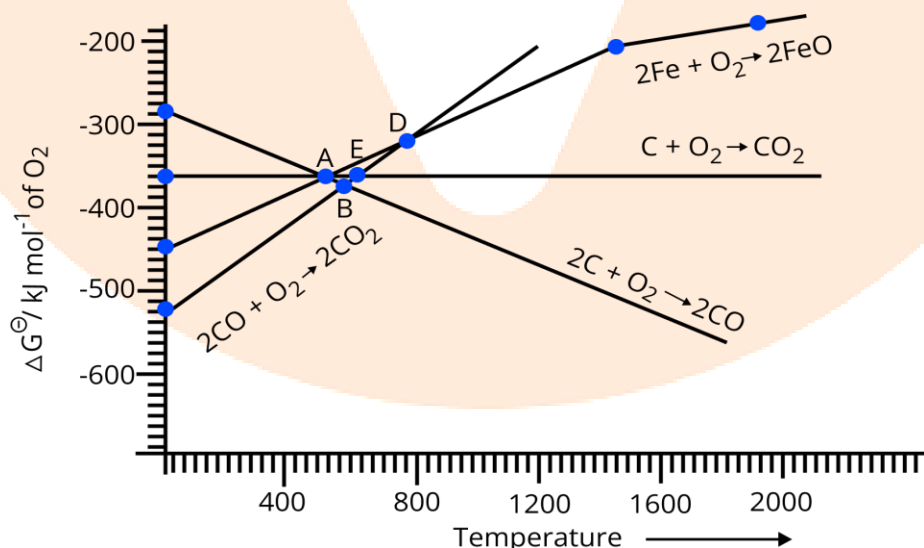
first step - The finely grounded ore of gold and silver are made to come in contact with the solution containing the cyanide,

Second step - it involves separation of gold and silver from the cyanide solution and

third step - it involves the recovery of gold and silver in their pure forms precipitating the remaining solution with zinc dust.

Thus, the metal is recovered by displacing Zn with the metal (Au or Ag) from metal ions.

Note: Answer the questions 11-13 on the basis of Fig. 6.1.



11. Choose the correct option of temperature at which carbon reduces FeO to iron and produces CO

- (i) Below temperature at point A.**
- (ii) Approximately at the temperature corresponding to point A.**
- (iii) Above temperature at point A but below the temperature at point D.**
- (iv) Above temperature at point A.**

Ans: Option (iv)

Above point A, $\Delta G_{(C,CO)} < \Delta G_{(Fe,FeO)}$. Thus, carbon reduces FeO to iron i.e. Fe and in order carbon is oxidized to carbon monoxide i.e. CO. Both these lines intersect at Point A. As we can see in the figure above point A only carbon could reduce FeO in Fe and CO is being produced.

12. Below point 'A' FeO can

- (i) be reduced by carbon monoxide only.**
- (ii) be reduced by both carbon monoxide and carbon.**
- (iii) be reduced by carbon only.**
- (iv) not be reduced by both carbon and carbon monoxide.**

Ans: Option (i)

Below point A, only the value of $\Delta G_{(CO,CO_2)}$ is less than the value of $\Delta G_{(Fe,FeO)}$ at the corresponding temperatures. Thus, only carbon monoxide will be able to reduce FeO to Fe and will get itself oxidized into CO_2 .

13. For the reduction of FeO at the temperature corresponding to point D, which of the following statements is correct?

- (i) ΔG° value for the overall reduction reaction with carbon monoxide is zero.**
- (ii) ΔG° value for the overall reduction reaction with a mixture of 1 mol carbon and 1mol oxygen is positive.**
- (iii) ΔG° value for the overall reduction reaction with a mixture of 2 mol carbon and 1mol oxygen will be positive.**
- (iv) ΔG° value for the overall reduction reaction with carbon monoxide is negative.**

Ans: Option (i)

According to the above graph, at point D the equivalent value of ΔG for the reduction of FeO is approximately 330 units for the particular temperature. Also, for CO the equivalent value of ΔG at the particular temperature for point D is around -330 units. Thus, if we calculate the overall value of ΔG for the reduction of FeO with carbon monoxide then it is almost zero at point D.

Multiple Choice Questions (Type II)

Note: In the following questions two or more options may be correct.

14. At the temperature corresponding to which of the points in Fig.6.1, FeO will be reduced to Fe by coupling the reaction $2\text{FeO} \rightarrow 2\text{Fe} + \text{O}_2$ with all of the following reactions?

(a) $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$ (b) $2\text{C} + \text{O}_2 \rightarrow 2\text{CO}$ and (c) $2\text{CO} + \text{O}_2 \rightarrow 2\text{CO}_2$.

- (i) Point A**
- (ii) Point B**
- (iii) Point D**

(iv) Point E

Ans: Option (ii) and (iv)

From Fig.6.1, at point B, all three lines, i.e. for the lines for all three given reactions, are below that of the line of reduction of FeO. This means, that the value of ΔG for the equivalent temperature at point B for reduction of FeO is greater than (positive) that of values of ΔG for all three reactions (negative). Thus, FeO will get reduced by all three reactions at point B. Same is the case with point E as that of point B, the value of ΔG for reduction of FeO at E is greater than the value of ΔG for all three reactions at the equivalent temperature. Thus, FeO will get reduced by all three reactions at point E.

15. Which of the following options are correct?

(i) Cast iron is obtained by smelting pig iron with scrap iron and coke using hot air blast.

(ii) In extraction of silver, silver is extracted as a cationic complex.

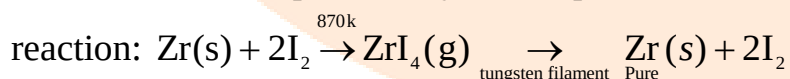
(iii) Nickel is purified by zone refining.

(iv) Zr and Ti are purified by van Arkel method.

Ans: Option (i) and (iv)

Pig iron is the iron obtained from the blast that has many impurities (Mn, P, Si, etc) and approx. 4% carbon content. When it is re-melted or cast with the help of scrap iron, then it has less carbon content i.e. 3% as compared to pig iron.

The Zr and Ti are purified by the help of Arkel Method that contain the following



Here we obtain Zr and Ti as a pure form of solid when purified using the Arkel method.

16. In the extraction of aluminium by Hall-Heroult process, purified Al_2O_3 is mixed with CaF_2 to

- (i) lower the melting point of Al_2O_3 .**
- (ii) increase the conductivity of molten mixture.**
- (iii) reduce Al^{3+} into Al(s) .**
- (iv) acts as a catalyst.**

Ans: Option (i) and (ii)

In the extraction of aluminium by Hall-Heroult's process, alumina is mixed with cryolite with the mixture of fluorspar & fluoride that increases the conductivity of electrical and lowers the melting point of Al_2O_3 . Hence option (i) and option (ii) are correct.

17. Which of the following statements is correct about the role of substances added in the froth floatation process?

- (i) Collectors enhance the non-wettability of the mineral particles.**
- (ii) Collectors enhance the wettability of gangue particles.**
- (iii) By using depressants in the process two sulphide ores can be separated.**
- (iv) Froth stabilisers decrease wettability of the gangue.**

Ans: Option (i) and (iii)

A suspension of powdered ore is made with water during the froth floatation process. Collectors and froth stabilisers are added to it. Collectors (e.g., pine oils, fatty acids, xanthates, etc.) improve the non wettability of the mineral particles, while froth stabilisers (e.g., cresols, aniline) keep the froth stable

18. In the Froth Floatation process, zinc sulphide and lead sulphide can be separated by

- (i) using collectors.**
- (ii) adjusting the proportion of oil to water.**
- (iii) using depression.**
- (iv) using froth stabilisers.**

Ans: Option (ii) and (iii)

It is possible to separate two sulphide ores using the froth floatation process by adjusting the proportion of oil to water or by using 'depressants.' In the case of an ore containing ZnS and PbS, for example, the depressant used is NaCN.

19. Common impurities present in bauxite are

- (i) CuO**
- (ii) ZnO**
- (iii) Fe_2O_3**
- (iv) SiO_2**

Ans: Option (iii) and (iv)

Bauxite consists of many impurities like hematite, goethite, Fe_2O_3 , the sand SiO_2 , etc.'

20. Which of the following ores are concentrated by froth floatation?

(i) Haematite

(ii) Galena

(iii) Copper pyrites

(iv) Magnetite

Ans: Option (ii) and (iii)

The froth floatation method is most commonly used for sulphide ore. Sulphide ores galena (PbS) and copper pyrites (CuFeS_2) are found here.

21. Which of the following reactions occur during calcination?}

(i) $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$

(ii) $2\text{FeS}_2 + \frac{11}{2}\text{O}_2 \rightarrow \text{Fe}_2\text{O}_3 + 4\text{SO}_2$

(iii) $\text{Al}_2\text{O}_3 \times x\text{H}_2\text{O} \rightarrow \text{Al}_2\text{O}_3 + x\text{H}_2\text{O}$

(iv) $\text{ZnS} + \frac{3}{2}\text{O}_2 \rightarrow \text{ZnO} + \text{SO}_2$

Ans: Option (i) and (iii)

Calcination is the process of heating when the volatile matter escapes, leaving the metal oxide behind. It is usually done in the absence of air.

22. For the metallurgical process of which of the ores calcined ore can be reduced by carbon?

(i) haematite

(ii) calamine

(iii) iron pyrites

(iv) sphalerite

Ans: Option (i) and (iii)

Explanation: Haematite is an ore of iron that can be calcined and reduced by carbon. In the metallurgical process, calamine ore is calcined ore that can be reduced by carbon. Hence, option (i) and (ii) are correct.

23. The main reactions occurring in blast furnaces during extraction of iron from haematite are



Ans: Option (i) and (iv)

(i) Carbon monoxide is the primary reducing agent in the furnace.

(ii) This is an endothermic reaction in which heat is absorbed from the furnace. As a result, it is critical not to add too much limestone, as this will cool the furnace. Calcium oxide is a basic oxide that reacts with acidic oxides in the rock, such as silicon dioxide. Calcium silicate is formed when calcium oxide reacts with silicon dioxide.

24. In which of the following methods of purification, metal is converted to its volatile compound which is decomposed to give pure metal?

(i) heating with a stream of carbon monoxide.

(ii) heating with iodine.

(iii) liquation.

(iv) distillation.

Ans: Option (i) and (ii)

In the vapour phase refining method, the metal is converted into its volatile compound and then is collected elsewhere. This involves two techniques:

1. Mond Process for refining Nickel: in this process, a volatile complex, nickel tetracarbonyl is formed when nickel is heated with a stream of carbon monoxide.

2. van Arkel Method for refining Zirconium or Titanium: this method is basically used for removal of oxygen and nitrogen present as impurities in the Zr or Ti metal. These are heated in an evacuated metal with iodine. As iodine is more covalent than these metals, it volatilizes.

25. Which of the following statements are correct?

(i) A depressant prevents a certain type of particle from coming to the froth.

(ii) Copper matte contains Cu_2S and ZnS .

(iii) The solidified copper obtained from reverberatory furnace has blistered appearance due to evolution of SO_2 during the extraction.

(iv) Zinc can be extracted by self-reduction.

Ans: Option (i) and (ii)

Explanation: Depressants are materials that are added for the separation of ores that prevent certain types of particles from coming to froth and forming bubbles. For example, an ore containing ZnS and PbS, NaCN is used as a depressant.

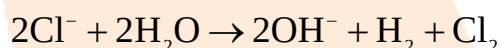
When sulphur ores are blown in hot air along with silica, the solidified metal which is obtained has a blistered appearance due to SO_2 evolution.

26. In the extraction of chlorine from brine

- (i) ΔG° for the overall reaction is negative.
- (ii) ΔG° for the overall reaction is positive.
- (iii) E° for overall reaction has negative value.
- (iv) E° for overall reaction has positive value.

Ans: Option (ii) and (iii)

Explanation: Using oxidation method for extraction of chlorine from brine. The reactions involved are:

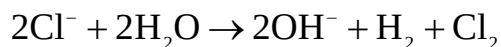


For this reaction, the value of $\Delta G^\circ = +422\text{kJ}$, which is positive. Using the formula $\Delta G^\circ = -nE^\circ F$, we get negative value of $E^\circ = -2.2\text{V}$.

Short Answer Type

27. Why is an external emf of more than 2.2 V required for the extraction of Cl_2 from the brine?

Ans: Extraction of chlorine from brine is an oxidation method. The reactions involved are:

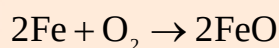
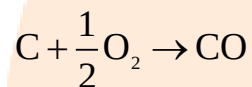


For this reaction, the value of $\Delta G^\circ = +422\text{kJ}$, which is positive. Using the formula, $\Delta G^\circ = -nE^\circ F$, we get negative value of $E^\circ = -2.2\text{V}$.

Since, $E^\circ = -2.2\text{V}$, it would naturally require an external voltage greater than 2.2V for the reaction to occur.

28. At temperatures above 1073K coke can be used to reduce FeO to Fe. How can you justify this reduction with the Ellingham diagram?

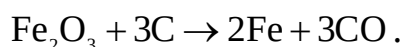
Ans: From Ellingham diagram, we know that $\Delta G^\circ_{(\text{C},\text{CO})} < \Delta G^\circ_{(\text{Fe},\text{Fe})}$, the following reactions are:



Therefore, FeO can be reduced to Fe by coke.

29. Wrought iron is the purest form of iron. Write a reaction used for the preparation of wrought iron from cast iron. How can the impurities of sulphur, silicon and phosphorus be removed from cast iron?

Ans: Iron oxides are mixed with limestone and coke in order to decompose carbonates and oxidize sulphides. This mixture is then fed into the blast furnace. Many reactions take place in different temperature ranges of the blast furnace. By oxidizing impurities from cast iron in a reverberatory furnace which is lined by hematite, wrought iron or malleable iron is obtained. This is the purest form of commercial iron. This can be shown with the following reaction:



Sulphur, silicone and phosphorus are oxidized and passed into the slag and removed as impurities when limestone is added as a flux to it.

30. How is copper extracted from low grade copper ores?

Ans: Low-grade copper ores contain a small percentage of copper that is 0.27% copper. By hydrometallurgy, low-grade copper is leached by treating with acid when copper metal goes into solution as copper ions. The solution containing copper ions is treated with scrap iron or hydrogen gas. Since iron or hydrogen is more reactive than copper so they reduced copper ions from solution to copper metal.

31. Write two basic requirements for refining of a metal by Mond process and by Van Arkel Method.

Ans: Both processes have two basic requirements:

- (i) The metal should react with an available reagent to form a volatile compound.
- (ii) The volatile compound should be easily decomposable, allowing for easy metal recovery.

32. Although carbon and hydrogen are better reducing agents but they are not used to reduce metallic oxides at high temperatures. Why?

Ans: The reduction is the removal of oxygen. Hydrogen and carbon both form oxides that are easy to remove both water and carbon dioxide which are produced as a gas at hot reactions. Although carbon and hydrogen are better reducing agents, they are not used to reduce metallic oxides at high temperatures because at high temperatures carbon and hydrogen react with metals to form carbide and hydrides respectively.

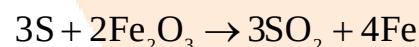
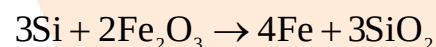
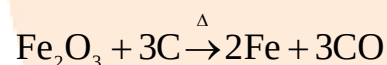
33. How do we separate two sulphide ores by the Froth Floatation Method? Explain with an example.

Ans: Separation of two sulphide ores can be accomplished by adjusting the proportion of oil to water or by using depressants. In the case of an ore containing ZnS and PbS, for example, the depressant NaCN is used. It forms a complex with ZnS and prevents it from coming into contact with froth, but PbS remains in contact with froth.

34. The purest form of iron is prepared by oxidising impurities from cast iron in a reverberatory furnace. Which iron ore is used to line the furnace? Explain by giving a reaction.

Ans: A metal is called Pure when it contains the highest concentration of itself and the least concentration of impurities. While refining of metal, these impurities are removed by various processes aided by the use of other compounds.

Hematite supplies oxygen and oxidizes carbon, silicon, manganese, and phosphorus present in the cast iron to carbon monoxide, manganese oxide, phosphorus pentoxide respectively.



Wrought iron is called the purest form of iron as it contains the highest percentage of Iron and the least impurities. It is prepared from cast iron by oxidizing impurities in a reverberatory furnace lined with Hematite.

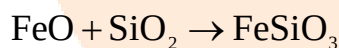
35. The mixture of compounds A and B is passed through a column of Al_2O_3 by using alcohol as eluent. Compound A is eluted in preference to compound B. Which of the compounds A or B, is more readily adsorbed on the column?

Ans: Column chromatography is a technique used to separate mixtures based on differences in their affinity towards the stationary phase. If one molecule in a solute has more affinity towards the stationary phase, it would form bonds with the stationary phase and hence remain in the column, Whereas, if the other one has less affinity, it would not form bonds and hence will travel down and come out of the column faster.

As compound A is eluted in preference to compound B, compound B is more readily adsorbed on the column of Al_2O_3 .

36: Why is sulphide ore of copper heated in a furnace after mixing with silica?

Ans: Sulphide ores are not reduced easily, but oxide ores are easily reduced. Sulphide ore of copper contains some types of impurities. This can cause by-product formation and degrade our final product. Iron oxide is present as a key impurity of concern in sulphide ore of copper. During the roasting process the temperature in the furnace is near about 1200°C . During such high temperatures iron oxide which is gangue (impurity) in ore forms slag. Slag is insoluble in molten metal and being lighter floats over the surface of molten metal. Here silica is used as flux.



37. Why are sulphide ores converted to oxide before reduction?

Ans: Sulphide ore of copper contains some types of impurities. Sulphide ores are not reduced easily, but oxide ores are easily reduced. It is necessary to make sulphide ores free from volatile impurities. This can be performed by a roasting method. In this moisture escape and impurities like sulphur, phosphorus, arsenic are oxidized to

their volatile oxides. It is performed in a reverberatory furnace. After this process the mass becomes porous.

38. Which method is used for refining Zr and Ti? Explain with equations.

Ans: Vapor phase refining or Van Arkel's method is used for refining Zr and Ti.

Vapour refining or vapour phase process: The metal is collected in the form of volatile compound that decomposes to pure metal.

Van Arkel method: This method is useful for removing all Oxygen and nitrogen present in the form of impurity in metals like Zr and Ti. The crude metal is heated in an empty vessel with a small amount of iodine. The metal iodine being more covalent volatilises. Impure Zr reacts with iodine to form zirconium tetra iodide.

The following reaction occurs $\text{Zr} + 2\text{I}_2 \rightarrow \text{ZrI}_4$ The metal iodide is decomposed on a tungsten filament electrically heated to about 1800 K. The pure metal is deposited on the filament and to maintain temperature the current is steadily raised, as the depositing continues.

39. What should be the considerations during the extraction of metals by electrochemical method?

Ans: The considerations during the extraction of metals by electrochemical method are :

a) Reactivity of metals: If the metals are reactive and are likely to react with water then the metals should be isolated by the electrolysis of their purified molten ore rather than their aqueous solution.

b) Suitability of electrodes: The electrode is carefully chosen so that they should not react with the product of electrolysis. If they react then the electrodes must be made of a material which is quite cheap so their replacement should not increase the cost of the process.

40. What is the role of flux in metallurgical processes?

Ans: In metallurgical processes, Flux works as various types of agents like chemical agents, cleaning/purifying agents.

Fluxes have many important properties such as corrosivity, cleanability, conductivity, volatility, etc.

Flux, when combined with gangue together, forms slag i.e. $\text{Flux} + \text{Gangue} = \text{Slag}$

As compared to gangue, slag separates much easily from the ore, in this way the removal of gangue also becomes much easier.

Flux is useful for making the molten mass more conducting which helps to remove impurities from metals through electrolytic metallurgical processes.

41. How are metals used as semiconductors refined? What is the principle of the method used?

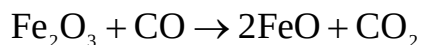
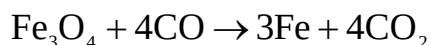
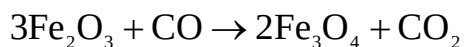
Ans: Semiconducting metal is produced by the zone refining method.

Zone refining is centered on the principle that the impurities which are included in the metal ores are more soluble in the molten state as compared to the solid-state of metals.

Zone refining is mostly used for obtaining semiconductor and other metals which are highly pure, e.g., germanium, silicon, etc.

42. Write down the reactions taking place in Blast furnaces related to the metallurgy of iron in the temperature range 500-800 K.

Ans: The chemical reactions taking place in Blast furnace in the temperature range 500-800 K are:



43. Give two requirements for vapour phase refining.

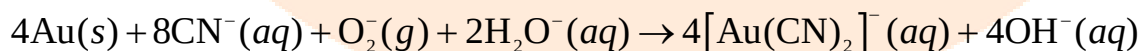
Ans: The two important requirements for performing the Vapour Phase Refining Method:

- The metal which is to be refined should form a volatile compound with an available reagent.
- The volatile compound formed must be easily decomposable so that it is easy to recover the desired metal.

44. Write the chemical reactions involved in the extraction of gold by cyanide process. Also give the role of zinc in the extraction.

Ans: The metal gold is a nonreactive metal. It is found as a native metal in its free state.

So, it is not compulsory to separate it by a chemical process. Thus, gold is leached to get it in its pure form. In the metallurgy of gold (Au), the respective metal is leached with a dilute mixture of NaCN or KCN in the presence of air where Gold(Au) is oxidized by oxygen of the air to Au^+ cation which then merges with ions to form a soluble compound.



Gold is then obtained from this soluble compound x by replacement method by using a much more electropositive Zinc metal.

In this method of Gold extraction, the metal Zinc (Zn) plays the role of a reducing agent and it reduces Au^+ to Au .

Zn itself gets oxidized to Zn^+ ions which combine with CN^- ions to form another soluble complex. $2\text{Na}[\text{Au}(\text{CN})_2](\text{aq}) + \text{Zn} \rightarrow 2\text{Au} + \text{Na}_2[\text{Zn}(\text{CN})_4]$.

Matching Type

45. Match the items of Column I with items of Column II and assign the correct code:

Column I	Column II
(A) Pendulum	(1) Chrome steel
(B) Malachite	(2) Nickel steel
(C) Calamine	(3) Na_3AlF_6
(D) Cryolite	(4) $\text{CuCO}_3 \times \text{Cu}(\text{OH})_2$
	(5) ZnCO_3

Code:

- (i) A (1) B (2) C (3) D (4)
- (ii) A (2) B (4) C (5) D (3)
- (iii) A (2) B (3) C (4) D (5)
- (iv) A (4) B (5) C (3) D (2)

Ans: Option (ii) A (2) B (4) C (5) D (3)

Pendulum is always made of nickel steel.

Hence, option (A) from column I is matched with option (2) from column II. Malachite is the ore of copper.

Hence, option (B) from column I is matched with option (4) from column II. Calamine is the ore of zinc.

Hence, option (C) from column I is matched with option (5) from column II. Cryolite is an ore of aluminium.

Hence, option (D) from column I is matched with option (3) from column II.

46. Match the items of Column I with the items of Column II and assign the correct code:

Column I	Column II
(A) Coloured bands	(1) Zone refining
(B) Impure metal to volatile complex	(2) Fractional distillation
(C) Purification of Ge and Si	(3) Mond process
(D) Purification of mercury	(4) Chromatography
	(5) Liquation

(i) A(1)B(2)C(4)D(5)

(ii) A(4)B(3)C(1)D(2)

(iii) A(3)B(4)C(2)D(1)

(iv) A(5)B(4)C(3)D(2)

Ans: Option (ii) A (4) B (3) C (1) D (2)

Coloured bands are found in chromatography.

Hence, option (A) from column I is matched with option (4) from column II. Impure metals are converted to volatile complexes in Mond's process.

Hence, option (B) from column I is matched with option (3) from column II. Purification of Ge and silicon is done using zone refining.

Hence, option (C) from column I is matched with option (1) from column II. Purification of mercury is done using fractional distillation.

Hence, option (D) from column I is matched with option (2) from column II.

47. Match items of Column I with the items of Column II and assign the correct code:

Column I	Column II
(A) Cyanide process	(1) Ultrapure Ge
(B) Froth Floatation Process	(2) Dressing of ZnS
(C) Electrolytic reduction	(3) Extraction of Al
(D) Zone refining	(4) Extraction of Au
	(5) Purification of Ni

(i) A(4) B(2) C(3) D(1)

(ii) A(2) B(3) C(1) D(5)

(iii) A(1) B(2) C(3) D(4)

(iv) A(3) B(4) C(5) D(1)

Ans: Correct code (i) A (4) B (2) C (3) D (1)

The cyanide process is used in the extraction of Au.

Hence, option (A) from column I is matched with option (4) from column II. Froth floatation process is used in the dressing of ZnS.

Hence, option (B) from column I is matched with option (2) from column II. Electrolytic reduction is used in the extraction of Al.

Hence, option (C) from column I is matched with option (3) from column II. Zone refining is used to get ultrapure Ge.

Hence, option (D) from column I is matched with option (1) from column II.

48. Match the items of Column I with the items of Column II and assign the correct code:

Column I	Column II
(A) Sapphire	(1) Al_2O_3
(B) Sphalerite	(2) NaCN
(C) Depressant	(3) Co
(D) Corundum	(4) ZnS
	(5) Fe_2O_3

(i) A(3)B(4)C(2)D(1)

(ii) A(5)B(4)C(3)D(2)

(iii) A(2)B(3)C(4)D(5)

(iv) A(1)B(2)C(3)D(4)

Ans: Option (i) A (3) B (4) C (2) D (1)

Sapphire is a gemstone containing Co.

Hence, option (A) from column I is matched with option (3) from column II. The Sphalerite single is ZnS.

Hence, option (B) from column I is matched with option (4) from column II. NaCN is also used as a depressant.

Hence, option (C) from column I is matched with option (2) from column II. Al_2O_3 is also called corundum.

Hence, option (D) from column I is matched with option (1) from column II.

49: Match the items of Column I with items of Column II and assign the correct code:

Column I	Column II
(A) Blistered Cu	(1) Aluminium
(B) Blast furnace	(2) $2\text{Cu}_2\text{O} + \text{Cu}_2\text{S} \rightarrow 6\text{Cu} + \text{SO}_2$
(C) Reverberatory furnace	(3) Iron
(D) Hall-Heroult process	(4) $\text{FeO} + \text{SiO}_2 \rightarrow \text{FeSiO}_3$
	(5) $2\text{Cu}_2\text{S} + 3\text{O}_2 \rightarrow 2\text{Cu}_2\text{O} + 2\text{SO}_2$

Code:

(i) A (2) B (3) C (4) D (1)

(ii) A (1) B (2) C (3) D (5)

(iii) A (5) B (4) C (3) D (2)

(iv) A (4) B (5) C (3) D (2)

Ans: Option (i) A (2) B (3) C (4) D (1)

Explanation:

A solidified copper has a blistered appearance due to the evolution of SO_2 . Hence it is called blistered copper.

Hence, option (A) from column I is matched with option (2) from column II.

Iron is extracted from a blast furnace.

Hence, option (B) from column I is matched with option (3) from column II.

The iron ore is heated in the reverberatory furnace after mixing with silica. In the furnace, iron oxide slags of iron and copper are produced in the form of copper matte.

Hence, option (C) from column I is matched with option (4) from column II.

The hall-Heroult process is used for the extraction of aluminium from its ore.

Hence, option (D) from column I is matched with option (1) from column II.

Assertion and Reason Type

In the following questions a statement of assertion followed by a statement of reason is given. Choose the correct answer out of the following choices.

- (i) Both assertion and reason are true and reason is the correct explanation of assertion.
- (ii) Both assertion and reason are true but reason is not the correct explanation of assertion.
- (iii) Assertion is true but the reason is false.
- (iv) Assertion is false but the reason is true.
- (v) Assertion and reason both are wrong.

50. Assertion: Nickel can be purified by the Mond process.

Reason: $\text{Ni}(\text{CO})_4$ is a volatile compound which decomposes at 460K to give pure Ni.

Ans: (i) Both assertion and reason are true and reason is the correct explanation of assertion.

Explanation: In the Mond process Nickel is reacted with carbon monoxide reversibly to give Nickel carbonyl, $\text{Ni}(\text{CO})_4$. Nickel carbonyl is a volatile compound. It decomposes to nickel and carbon monoxide at 460K.

51. Assertion: Zirconium can be purified by Van Arkel method.

Reason: ZrI_4 is volatile and decomposes at 1800K.

Ans: (i) Both assertion and reason are true and reason is the correct explanation of assertion.

Explanation: Van Arkel method is generally used to obtain pure forms of Zirconium (Zr) and Titanium (Ti).

The metal iodide is heated at 1800 K and is decomposed on a tungsten filament. The pure metal is dropped on the tungsten filament. This proves that ZrI_4 is volatile and decomposes at 1800 K.

52. Assertion: Sulphide ores are concentrated by Froth Flotation method.

Reason: Cresols stabilise the froth in the Froth Flotation method.

Ans: (ii) Both assertion and reason are true but reason is not the correct explanation of assertion. Explanation: Froth Flotation method is used to separate the hydrophobic materials from hydrophilic materials. In this method a mixture of palm oil, water and detergent is taken in a tank along with powdered sulphide ore. Compressed air is then passed through the pipe of the rotating agitator to create froth. The sulphide ore is then wetted by the palm oil mixture and it rises with the froth and the impurities or gauge settles at the bottom of the tank. The froth containing the sulphide is then cleaned and dried. Cresols or Aniline are used to stabilize the froth and in case of sulphides palm oil is used to collect non wet particles of the sulphide.

53. Assertion: Zone refining method is very useful for producing semiconductors.

Reason: Semiconductors are of high purity.

Ans: (ii) Both assertion and reason are true but reason is not the correct explanation of assertion.

Explanation: The zone refining method is very useful for producing high-purity semiconductors and other metals, such as germanium.

54. Assertion: Hydrometallurgy involves dissolving the ore in a suitable reagent followed by precipitation by a more electropositive metal.

Reason: Copper is extracted by hydrometallurgy.

Ans: (ii) Both assertion and reason are true but reason is not the correct explanation of assertion.

Explanation: Hydrometallurgy is used to extract copper from low-grade ore. Hydrometallurgy entails dissolving the ore in a suitable reagent and then precipitating it. In this method, more electropositive metal is used, allowing pure metal to be displaced.

Long Answer Type

55. Explain the following:

(a) CO_2 is a better reducing agent below 710K whereas CO is a better reducing agent above 710K.

Ans: According to Ellingham diagram, the reaction of CO_2 is more feasible at temperatures lower than 710K and thus it is a better reducing agent below 710K.

While the reaction of CO is more feasible at temperatures higher than 710K and thus it is a better reducing agent at above 710K.

(b) Generally, sulphide ores are converted into oxides before reduction.

Ans: According to the Ellingham diagram, the more negative the Gibbs free energy of a particular reaction the more feasible it is to carry out. Since the oxides are easier to reduce, sulphide ores are converted into oxides before reduction.

(c) Silica is added to the sulphide ore of copper in the reverberatory furnace.

Ans: To extract copper, its sulphide ores are supposed to be heated in a reverberatory furnace. The sulphide ore may contain iron as an impurity. So, it is mixed with silica

before heating. The oxide of iron slags off as iron silicate. In other words, the iron impurity is removed easily in the form of iron silicate.

(d) Carbon and hydrogen are not used as reducing agents at high temperatures.

Ans: Carbon and hydrogen are good reducing agents. They are used preferably for reduction of oxides. They also form escapable gases as by-products. However, at higher temperatures they form carbides and hydrides. Also, it is not convenient to maintain such high temperatures. Thus, electrolysis is used for this purpose.

(e) Vapour phase refining method is used for the purification of Ti.

Ans: Ti is heated in an evacuated vessel along with iodine. Ti forms volatile compounds with iodine. The volatile compound formed decomposes easily. This makes the recovery of the metal convenient. Ti thus fulfils the requirements needed for carrying out vapour phase refining. Hence, this method is used for its purification.